

INFO6007

Project Management in IT

Week 10: Integration Management

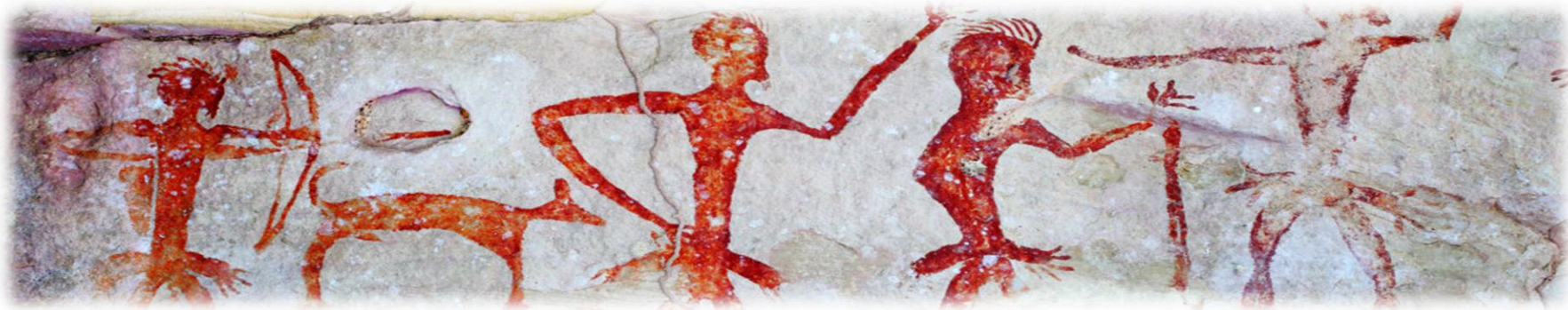
Semester 1, 2026



Acknowledgement of Country

I would like to acknowledge the Traditional Owners of Australia and recognise their continuing connection to land, water and culture. I am currently on the land of the Gadigal people of the Eora nation and pay my respects to their Elders, past, present and emerging.

I further acknowledge the Traditional Owners of the country on which you are on and pay respects to their Elders, past, present and future.



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Agenda

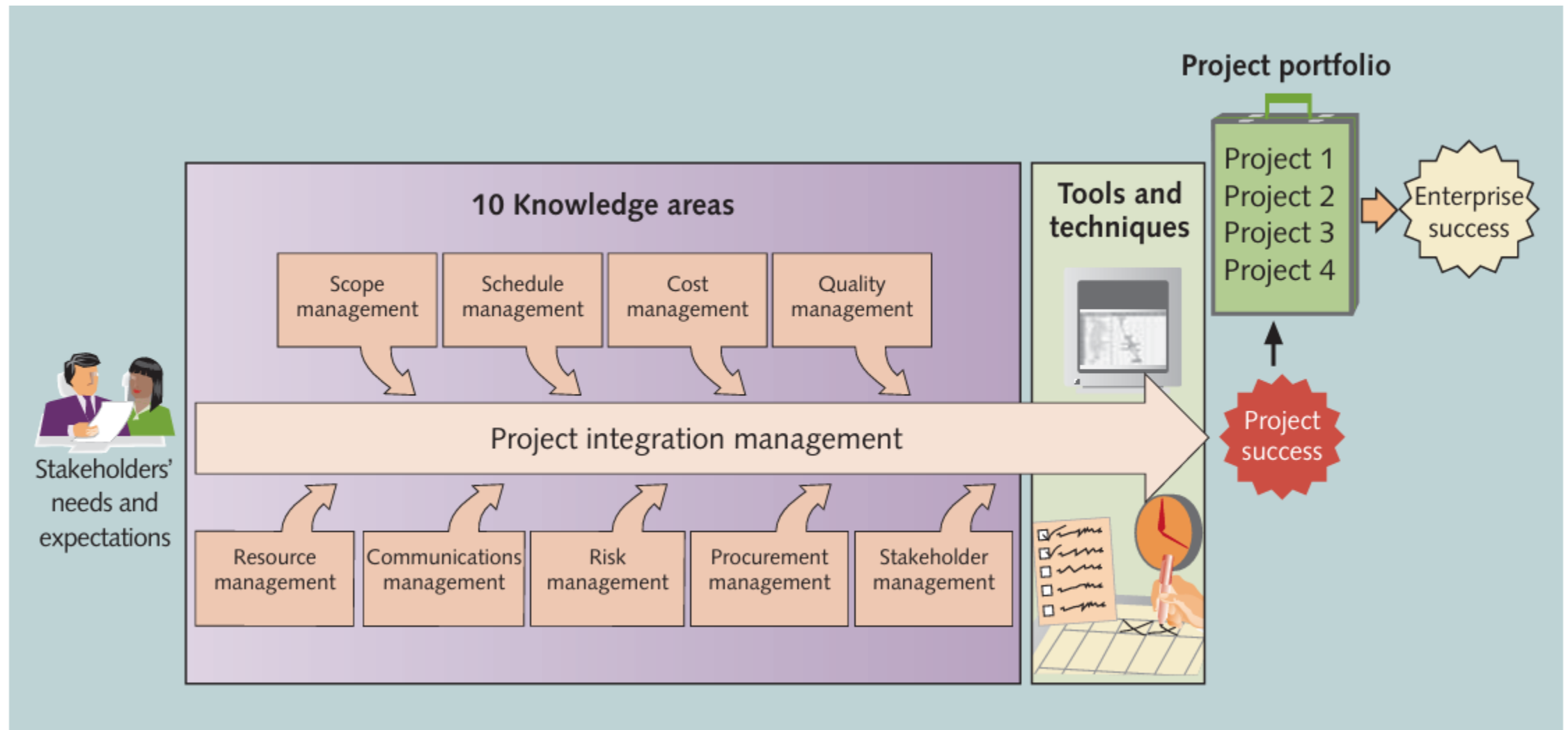
- Introduction
- Recap of the Knowledge areas
- Recap of the Project Lifecycle
- Uniqueness of IT projects
- Change management

Introduction

The PMBOK Framework

- **PMBOK** stands for **Project Management Body of Knowledge**, published by the Project Management Institute (PMI).
- It is a **globally recognized standard** for project management, providing a structured approach to managing projects across industries, including IT.
- The framework organizes project management into **5 Process Groups** and **10 Knowledge Areas**, providing guidance on **what to do** and **how to integrate processes** to achieve project success.

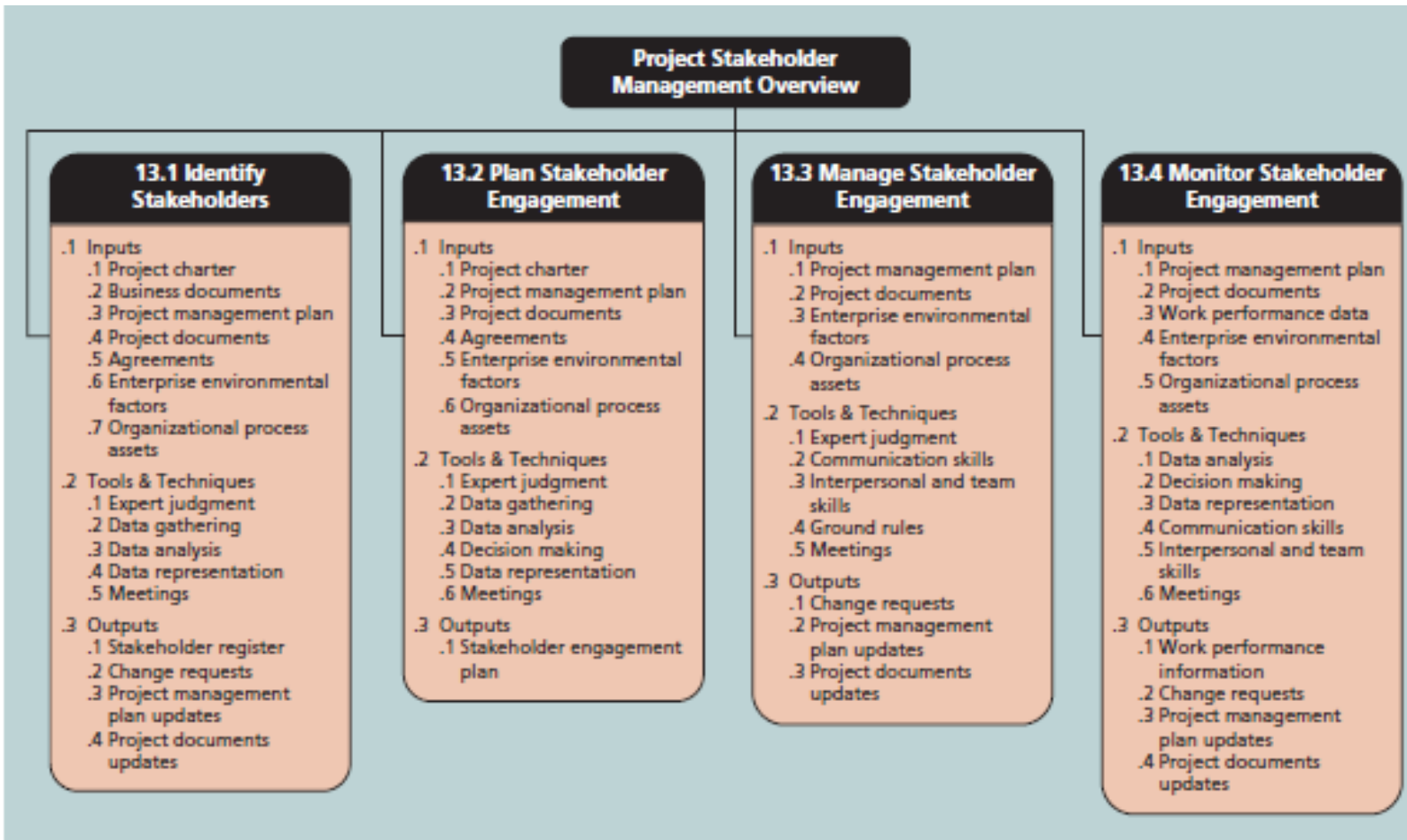
PMBOK Framework



Source: Project Management Institute, Inc., *A Guide to the Project Management Body of Knowledge (PMBOK® Guide) – Sixth Edition* (2017).

Recap of the Knowledge Areas

Knowledge Area 1 – Stakeholder Management



- **Purpose:** Identify, engage, and manage stakeholders to meet expectations.
- **Key activities:** Identify stakeholders, plan engagement, manage engagement, monitor engagement.
- **IT example:** Ensuring business users, IT teams, and vendors are aligned on system requirements.

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Project Scope Management Overview

5.1 Plan Scope Management

- .1 Inputs
 - .1 Project charter
 - .2 Project management plan
 - .3 Enterprise environmental factors
 - .4 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Data analysis
 - .3 Meetings
- .3 Outputs
 - .1 Scope management plan
 - .2 Requirements management plan

5.4 Create WBS

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Enterprise environmental factors
 - .4 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Decomposition
- .3 Outputs
 - .1 Scope baseline
 - .2 Project documents updates

5.2 Collect Requirements

- .1 Inputs
 - .1 Project charter
 - .2 Project management plan
 - .3 Project documents
 - .4 Business documents
 - .5 Agreements
 - .6 Enterprise environmental factors
 - .7 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Data gathering
 - .3 Data analysis
 - .4 Decision making
 - .5 Data representation
 - .6 Interpersonal and team skills
 - .7 Context diagram
 - .8 Prototypes
- .3 Outputs
 - .1 Requirements documentation
 - .2 Requirements traceability matrix

5.5 Validate Scope

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Verified deliverables
 - .4 Work performance data
- .2 Tools & Techniques
 - .1 Inspection
 - .2 Decision making
- .3 Outputs
 - .1 Accepted deliverables
 - .2 Work performance information
 - .3 Change requests
 - .4 Project documents updates

5.3 Define Scope

- .1 Inputs
 - .1 Project charter
 - .2 Project management plan
 - .3 Project documents
 - .4 Enterprise environmental factors
 - .5 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Data analysis
 - .3 Decision making
 - .4 Interpersonal and team skills
 - .5 Product analysis
- .3 Outputs
 - .1 Project scope statement
 - .2 Project documents updates

5.6 Control Scope

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Work performance data
 - .4 Organizational process assets
- .2 Tools & Techniques
 - .1 Data analysis
- .3 Outputs
 - .1 Work performance information
 - .2 Change requests
 - .3 Project management plan updates
 - .4 Project documents updates

Knowledge Area 2 - Scope Management

- **Purpose:** Defines what is included and excluded from the project.
- **Key activities:** Collect requirements, define scope, create WBS, validate and control scope.
- **IT example:** Determining features for a new software module without exceeding project boundaries.

Project Schedule Management Overview

6.1 Plan Schedule Management

- .1 Inputs
 - .1 Project charter
 - .2 Project management plan
 - .3 Enterprise environmental factors
 - .4 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Data analysis
 - .3 Meetings
- .3 Outputs
 - .1 Schedule management plan

6.4 Estimate Activity Durations

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Enterprise environmental factors
 - .4 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Analogous estimating
 - .3 Parametric estimating
 - .4 Three-point estimating
 - .5 Bottom-up estimating
 - .6 Data analysis
 - .7 Decision making
 - .8 Meetings
- .3 Outputs
 - .1 Duration estimates
 - .2 Basis of estimates
 - .3 Project documents updates

6.2 Define Activities

- .1 Inputs
 - .1 Project management plan
 - .2 Enterprise environmental factors
 - .3 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Decomposition
 - .3 Rolling wave planning
 - .4 Meetings
- .3 Outputs
 - .1 Activity list
 - .2 Activity attributes
 - .3 Milestone list
 - .4 Change requests
 - .5 Project management plan updates

6.5 Develop Schedule

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Agreements
 - .4 Enterprise environmental factors
 - .5 Organizational process assets
- .2 Tools & Techniques
 - .1 Schedule network analysis
 - .2 Critical path method
 - .3 Resource optimization
 - .4 Data analysis
 - .5 Leads and lags
 - .6 Schedule compression
 - .7 Project management information system
 - .8 Agile release planning
- .3 Outputs
 - .1 Schedule baseline
 - .2 Project schedule
 - .3 Schedule data
 - .4 Project calendars
 - .5 Change requests
 - .6 Project management plan updates
 - .7 Project documents updates

6.3 Sequence Activities

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Enterprise environmental factors
 - .4 Organizational process assets
- .2 Tools & Techniques
 - .1 Precedence diagramming method
 - .2 Dependency determination and integration
 - .3 Leads and lags
 - .4 Project management information system
- .3 Outputs
 - .1 Project schedule network diagrams
 - .2 Project documents updates

6.6 Control Schedule

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Work performance data
 - .4 Organizational process assets
- .2 Tools & Techniques
 - .1 Data analysis
 - .2 Critical path method
 - .3 Project management information system
 - .4 Resource optimization
 - .5 Leads and lags
 - .6 Schedule compression
- .3 Outputs
 - .1 Work performance information
 - .2 Schedule forecasts
 - .3 Change requests
 - .4 Project management plan updates
 - .5 Project documents updates

Knowledge Area 3 - Schedule Management

- **Purpose:** Plan and control project timelines to ensure on-time delivery.
- **Key activities:** Define activities, sequence, estimate durations, develop and control schedule.
- **IT example:** Scheduling sprints in agile development or milestones in a waterfall project.

Project Cost Management Overview

7.1 Plan Cost Management

- .1 Inputs
 - .1 Project charter
 - .2 Project management plan
 - .3 Enterprise environmental factors
 - .4 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Data analysis
 - .3 Meetings
- .3 Outputs
 - .1 Cost management plan

7.2 Estimate Costs

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Enterprise environmental factors
 - .4 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Analogous estimating
 - .3 Parametric estimating
 - .4 Bottom-up estimating
 - .5 Three-point estimating
 - .6 Data analysis
 - .7 Project management information system
 - .8 Decision making
- .3 Outputs
 - .1 Cost estimates
 - .2 Basis of estimates
 - .3 Project documents updates

7.3 Determine Budget

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Business documents
 - .4 Agreements
 - .5 Enterprise environmental factors
 - .6 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Cost aggregation
 - .3 Data analysis
 - .4 Historical information review
 - .5 Funding limit reconciliation
 - .6 Financing
- .3 Outputs
 - .1 Cost baseline
 - .2 Project funding requirements
 - .3 Project documents updates

7.4 Control Costs

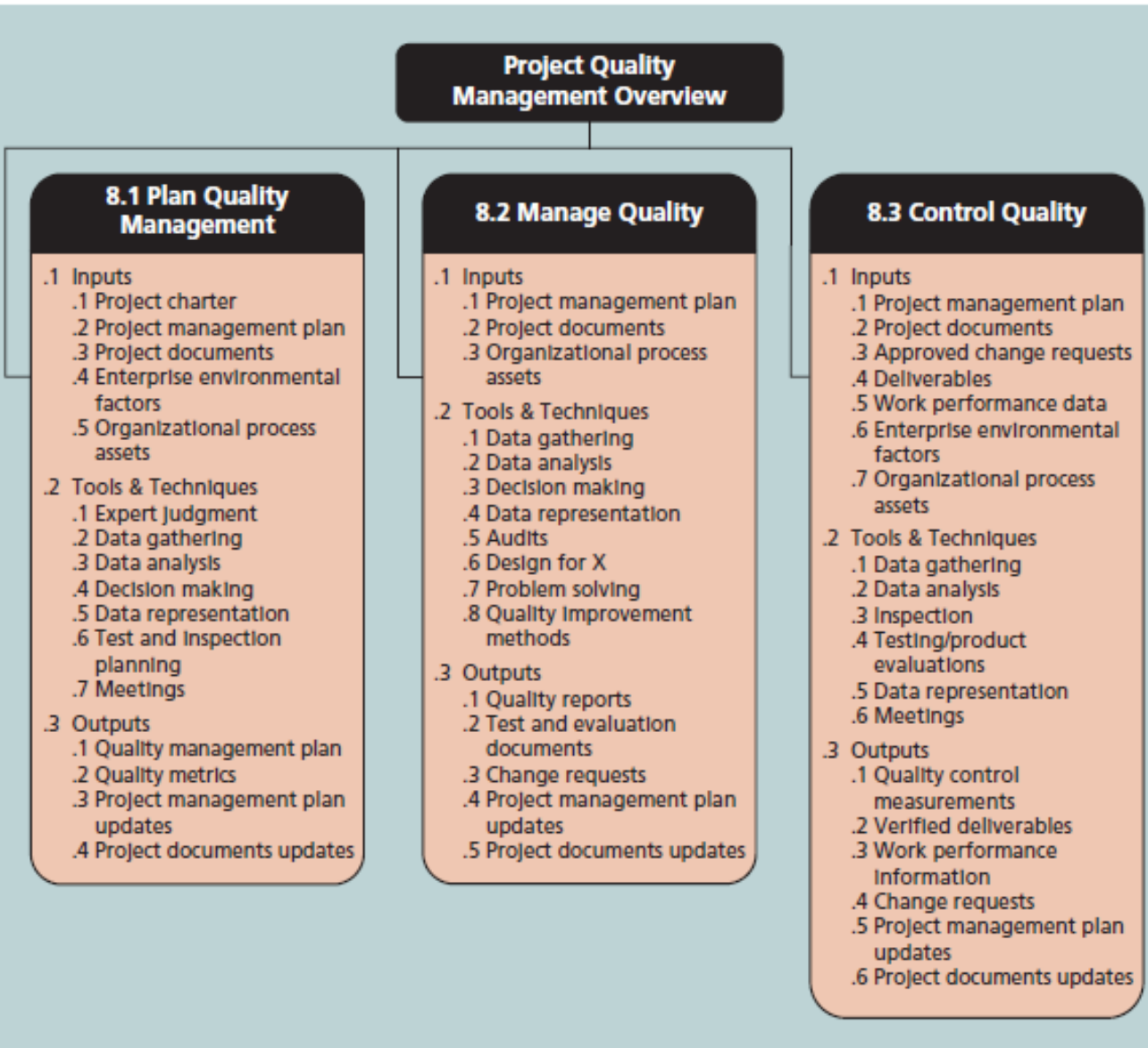
- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Project funding requirements
 - .4 Work performance data
 - .5 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Data analysis
 - .3 To-complete performance index
 - .4 Project management information system
- .3 Outputs
 - .1 Work performance information
 - .2 Cost forecasts
 - .3 Change requests
 - .4 Project management plan updates
 - .5 Project documents updates

Knowledge Area 4 - Cost Management

- **Purpose:** Plan, estimate, budget, and control project costs.
- **Key activities:** Estimate costs, determine budget, control costs to prevent overruns.
- **IT example:** Managing budget for cloud infrastructure, licensing, and developer resources.

Knowledge Area 5 - Quality Management

- **Purpose:** Ensure deliverables meet defined standards and satisfy stakeholders.
- **Key activities:** Plan quality, perform quality assurance, control quality.
- **IT example:** Ensuring software passes testing and meets security/compliance standards.



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Project Resource Management Overview

9.1 Plan Resource Management

- .1 Inputs
 - .1 Project charter
 - .2 Project management plan
 - .3 Project documents
 - .4 Enterprise environmental factors
 - .5 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Data representation
 - .3 Organizational theory
 - .4 Meetings
- .3 Outputs
 - .1 Resource management plan
 - .2 Team charter
 - .3 Project documents updates

9.4 Develop Team

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Enterprise environmental factors
 - .4 Organizational process assets
- .2 Tools & Techniques
 - .1 Colocation
 - .2 Virtual teams
 - .3 Communication technology
 - .4 Interpersonal and team skills
 - .5 Recognition and rewards
 - .6 Training
 - .7 Individual and team assessments
 - .8 Meetings
- .3 Outputs
 - .1 Team performance assessments
 - .2 Change requests
 - .3 Project management plan updates
 - .4 Project documents updates
 - .5 Enterprise environmental factors updates
 - .6 Organizational process assets updates

9.2 Estimate Activity Resources

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Enterprise environmental factors
 - .4 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Bottom-up estimating
 - .3 Analogous estimating
 - .4 Parametric estimating
 - .5 Data analysis
 - .6 Project management information system
 - .7 Meetings
- .3 Outputs
 - .1 Resource requirements
 - .2 Basis of estimates
 - .3 Resource breakdown structure
 - .4 Project documents updates

9.5 Manage Team

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Work performance reports
 - .4 Team performance assessments
 - .5 Enterprise environmental factors
 - .6 Organizational process assets
- .2 Tools & Techniques
 - .1 Interpersonal and team skills
 - .2 Project management information system
- .3 Outputs
 - .1 Change requests
 - .2 Project management plan updates
 - .3 Project documents updates
 - .4 Enterprise environmental factors updates

9.3 Acquire Resources

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Enterprise environmental factors
 - .4 Organizational process assets
- .2 Tools & Techniques
 - .1 Decision making
 - .2 Interpersonal and team skills
 - .3 Pre-assignment
 - .4 Virtual teams
- .3 Outputs
 - .1 Physical resource assignments
 - .2 Project team assignments
 - .3 Resource calendars
 - .4 Change requests
 - .5 Project management plan updates
 - .6 Project documents updates
 - .7 Enterprise environmental factors updates
 - .8 Organizational process assets updates

9.6 Control Resources

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Work performance data
 - .4 Agreements
 - .5 Organizational process assets
- .2 Tools & Techniques
 - .1 Data analysis
 - .2 Problem solving
 - .3 Interpersonal and team skills
 - .4 Project management information system
- .3 Outputs
 - .1 Work performance information
 - .2 Change requests
 - .3 Project management plan updates
 - .4 Project documents updates

Knowledge Area 6 - Resource Management

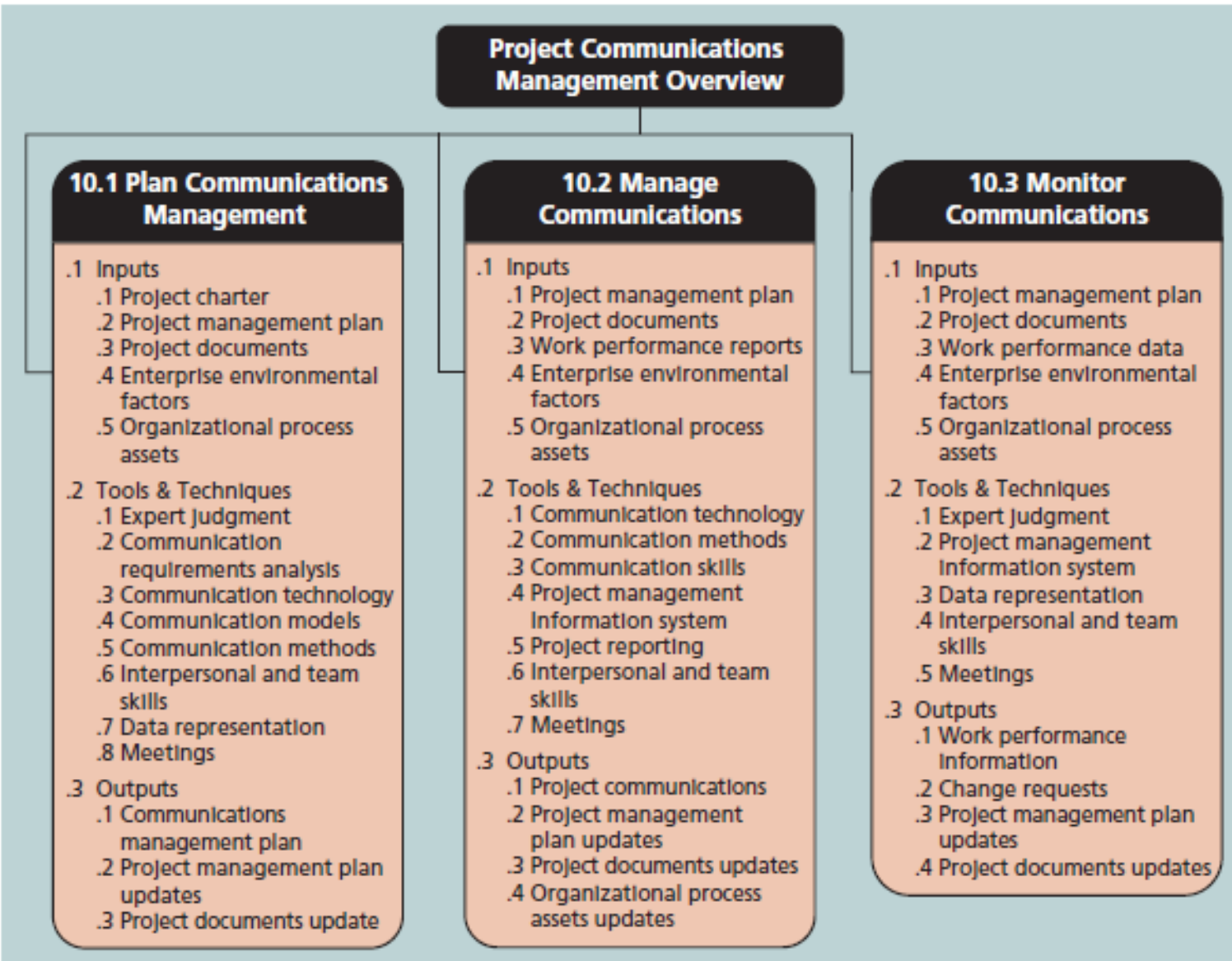
Purpose: Plan, acquire, and manage people, equipment, and materials.

Key activities: Resource planning, estimating, acquiring, developing, and managing teams.

IT example: Assigning developers, testers, and system admins efficiently.

Knowledge Area 7 - Communication Management

- **Purpose:** Ensure timely and effective information flow.
- **Key activities:** Plan communications, manage communications, monitor communications.
- **IT example:** Reporting progress to stakeholders, updating teams on sprint progress.



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Project Risk Management Overview

11.1 Plan Risk Management

- .1 Inputs
 - .1 Project charter
 - .2 Project management plan
 - .3 Project documents
 - .4 Enterprise environmental factors
 - .5 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Data analysis
 - .3 Meetings
- .3 Outputs
 - .1 Risk management plan

11.2 Identify Risks

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Agreements
 - .4 Procurement documentation
 - .5 Enterprise environmental factors
 - .6 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Data gathering
 - .3 Data analysis
 - .4 Interpersonal and team skills
 - .5 Prompt lists
 - .6 Meetings
- .3 Outputs
 - .1 Risk register
 - .2 Risk report
 - .3 Project documents updates

11.3 Perform Qualitative Risk Analysis

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Enterprise environmental factors
 - .4 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Data gathering
 - .3 Data analysis
 - .4 Interpersonal and team skills
 - .5 Risk categorization
 - .6 Data representation
 - .7 Meetings
- .3 Outputs
 - .1 Project documents updates

11.4 Perform Quantitative Risk Analysis

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Enterprise environmental factors
 - .4 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Data gathering
 - .3 Interpersonal and team skills
 - .4 Representations of uncertainty
 - .5 Data analysis
- .3. Outputs
 - .1 Project documents updates

11.5 Plan Risk Responses

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Enterprise environmental factors
 - .4 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Data gathering
 - .3 Interpersonal and team skills
 - .4 Strategies for threats
 - .5 Strategies for opportunities
 - .6 Contingent response strategies
 - .7 Strategies for overall project risk
 - .8 Data analysis
 - .9 Decision making
- .3 Outputs
 - .1 Change requests
 - .2 Project management plan updates
 - .3 Project documents updates

11.6 Implement Risk Responses

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Interpersonal and team skills
 - .3 Project management information system
- .3 Outputs
 - .1 Change requests
 - .2 Project documents updates

11.7 Monitor Risks

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Work performance data
 - .4 Work performance reports
- .2. Tools & Techniques
 - .1 Data analysis
 - .2 Audits
 - .3 Meetings
- .3. Outputs
 - .1 Work performance information
 - .2 Change requests
 - .3 Project management plan updates
 - .4 Project documents updates
 - .5 Organizational process assets updates

Knowledge Area 8 - Risk Management

- **Purpose:** Identify, analyze, and mitigate project risks.
- **Key activities:** Risk identification, qualitative & quantitative analysis, risk response planning, risk monitoring.
- **IT example:** Planning for potential server downtime, security breaches, or requirement changes.

Project Procurement Management Overview

12.1 Plan Procurement Management

- .1 Inputs
 - .1 Project charter
 - .2 Business documents
 - .3 Project management plan
 - .4 Project documents
 - .5 Enterprise environmental factors
 - .6 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Data gathering
 - .3 Data analysis
 - .4 Source selection analysis
 - .5 Meetings
- .3 Outputs
 - .1 Procurement management plan
 - .2 Procurement strategy
 - .3 Bid documents
 - .4 Procurement statement of work
 - .5 Source selection criteria
 - .6 Make-or-buy decisions
 - .7 Independent cost estimates
 - .8 Change requests
 - .9 Project documents updates
 - .10 Organizational process assets updates

12.2 Conduct Procurements

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Procurement documentation
 - .4 Seller proposals
 - .5 Enterprise environmental factors
 - .6 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Advertising
 - .3 Bidder conferences
 - .4 Data analysis
 - .5 Interpersonal and team skills
- .3 Outputs
 - .1 Selected sellers
 - .2 Agreements
 - .3 Change requests
 - .4 Project management plan updates
 - .5 Project documents updates
 - .6 Organizational process assets updates

12.3 Control Procurements

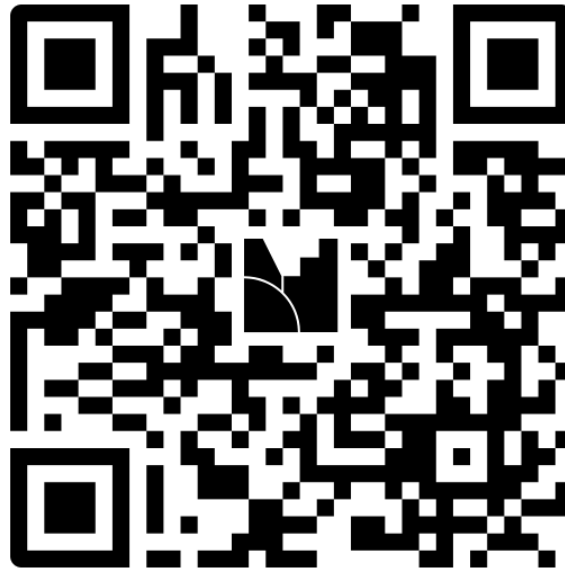
- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Agreements
 - .4 Procurement documentation
 - .5 Approved change requests
 - .6 Work performance data
 - .7 Enterprise environmental factors
 - .8 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Claims administration
 - .3 Data analysis
 - .4 Inspection
 - .5 Audits
- .3 Outputs
 - .1 Closed procurements
 - .2 Work performance information
 - .3 Procurement documentation updates
 - .4 Change requests
 - .5 Project management plan updates
 - .6 Project documents updates
 - .7 Organizational process assets updates

Knowledge Area 9 - Procurement Management

- **Purpose:** Acquire goods, services, or results from external sources.
- **Key activities:** Plan procurement, conduct procurement, control procurement, close procurement.
- **IT example:** Hiring cloud service providers or purchasing software licenses.

Menti: In your opinion what is the most important knowledge area in PMBOK? Why?

- <https://www.menti.com/alwzaz71hd97>



Recap of the Project Lifecycle

Project Lifecycle Phases

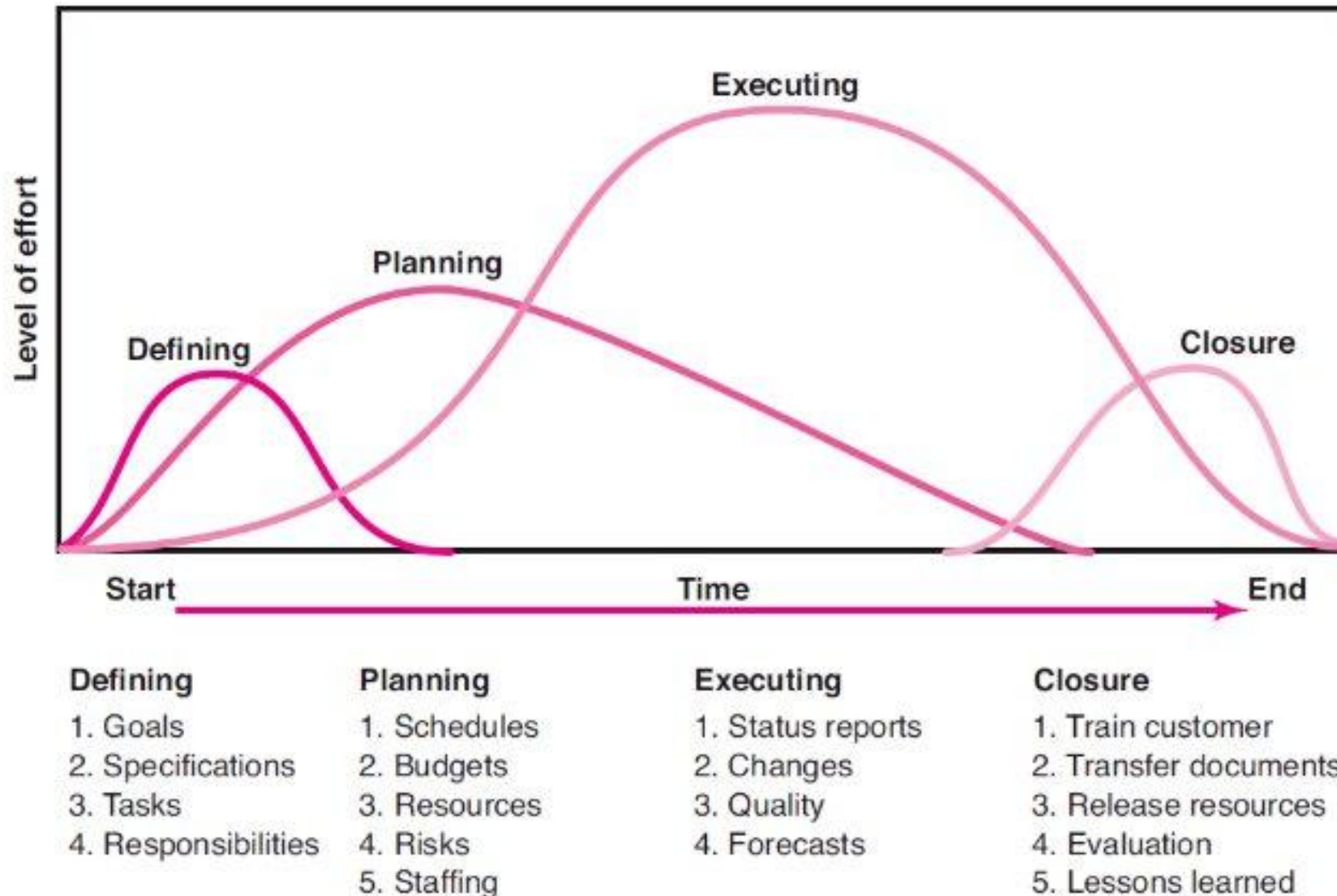


Figure 2.1: Project Life Cycle

The Initiation Stage

- **Purpose:** Formally start the project and define its objectives.
- **Key Activities:**
 - Develop Project Charter (formal authorization).
 - Identify key stakeholders.
 - Conduct feasibility study (technical, financial, operational).
- **Outputs:** Project charter, stakeholder register, preliminary scope.
- **Key Consideration:** Ensures the project has a clear purpose and sponsor support.

The Planning stage

- **Purpose:** Create a detailed roadmap to achieve project objectives.
- **Key Activities:**
 - Define **scope**, create **Work Breakdown Structure (WBS)**, Develop Schedule, Estimate costs and create a Budget
 - Plan **resources**, **risk management**, **communications**, and **procurement**.
 - Consolidate all plans into a **Project Management Plan**.
- **Outputs:** Scope statement, schedule, budget, risk plan, project management plan.
- **Key Consideration:** Good planning reduces uncertainties and sets realistic expectations.

The Executing Stage

- **Purpose:** Perform the work to deliver the project outputs.
- **Key Activities:**
 - Assign and manage resources.
 - Develop project deliverables (coding, configuring, testing).
 - Implement quality assurance and communication plans.
 - Manage stakeholder expectations.
- **Outputs:** Work performance data, completed deliverables.
- **Key Consideration:** Strong coordination and monitoring are needed to keep the project on track.

The Monitoring and Controlling stage

- **Purpose:** Track progress, identify deviations, and implement corrective actions.
- **Key Activities:**
 - Measure performance against scope, schedule, and cost baselines.
 - Perform integrated change control for scope, schedule, or cost changes.
 - Monitor risks, quality, resources, and communications.
- **Outputs:** Performance reports, approved change requests, updates to project plans.
- **Key Consideration:** This stage is continuous during execution and ensures the project stays aligned with objectives.

The Closure stage

- **Purpose:** Formally complete and close the project or project phase.
- **Key Activities:**
 - Obtain formal acceptance of deliverables.
 - Conduct lessons learned session.
 - Release project resources.
 - Archive project documentation.
- **Outputs:** Final project report, lessons learned, formal acceptance, closed contracts.
- **Key Consideration:** Proper closure ensures knowledge transfer and sets the stage for future projects.

Mapping the Knowledge Areas to Lifecycle Phases

	Initiating	Planning	Executing	Monitoring & Control	Closing
Scope Management		✓		✓	
Schedule Management		✓		✓	
Cost Management		✓		✓	
Quality Management		✓	✓	✓	
Resource Management		✓	✓	✓	
Communication Management	✓	✓	✓	✓	✓
Risk Management		✓	✓	✓	
Procurement Management		✓	✓	✓	✓
Stakeholder Management	✓	✓	✓	✓	✓
Integration Management	✓	✓	✓	✓	✓

Uniqueness of IT Projects

Characteristics of IT Projects

- **High level of complexity**

- IT projects often involve multiple interdependent systems, applications, and technologies.
- Complexity arises from integrating hardware, software, networks, and cloud services.
- Example: Implementing an enterprise ERP system requires coordination between finance, HR, and IT teams.

- **Rapidly changing technology**

- Technology evolves quickly, making IT projects more susceptible to obsolescence.
- Teams must adapt to new tools, programming languages, or platforms mid-project.
- Example: Switching from on-premise servers to cloud infrastructure during the project lifecycle.

Characteristics of IT Projects

- **Intangible deliverables**

- Unlike construction or manufacturing, IT projects often produce software, systems, or digital solutions, which are intangible.
- Success is measured by functionality, performance, and user satisfaction, not physical output.
- Example: A web application or mobile app that users cannot physically see until deployment.

- **High stakeholder involvement**

- IT projects require active participation from diverse stakeholders: developers, testers, operations, end-users, management.
- Misalignment can cause scope creep or failure.
- Example: Customizing a CRM system needs input from sales, marketing, and IT teams.

Characteristics of IT Projects

- **High uncertainty and risk**

- IT projects often face uncertainty in requirements, technology performance, or integration challenges.
- Risk management is critical throughout the project lifecycle.
- Example: Unanticipated software bugs delaying a product launch.

- **Dependency on requirements accuracy**

- IT projects are highly sensitive to requirements definition; small misunderstandings can cascade into significant delays or rework.
- Agile methodologies are often used to manage evolving requirements.
- Example: A mobile app with misunderstood user login flows may require major redesign if requirements were unclear.

What is Integration Management?

- **Project Integration Management** is the **coordination of all processes and activities across the project lifecycle** to ensure the project meets its objectives.
- It is sometimes called the “**glue**” that holds a project together.
- Key Ideas:
 - Every project knowledge area (scope, schedule, cost, quality, etc.) must be aligned and integrated.
 - Integration management ensures decisions, changes, and activities are coordinated, avoiding conflicts and duplication.

Integration in Practice

- Coordinating all project elements
 - Integration management ensures that **scope, schedule, cost, quality, resources, risks, and communications** work together seamlessly.
- Tools and Techniques in practice:
 - **Project Management Information Systems (PMIS):** e.g., MS Project, Jira, Asana, or Trello.
 - **Dashboards:** Real-time updates on progress, risks, costs, and resources.
 - **Change Control Boards (CCB):** Teams that approve/reject project changes.
 - **Documentation & Knowledge Repositories:** Wikis, Confluence, or SharePoint for lessons learned and best practices.

Change Control Boards (CCBs)



Real world Integration Processes

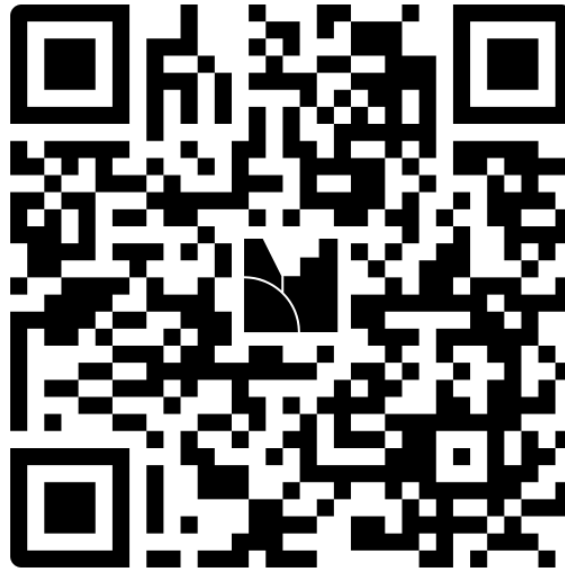
- Developing the Project Charter
 - Create the project initiation document with goals, objectives, stakeholders, and high-level requirements.
- Developing the Project Management Plan
 - Consolidate all subsidiary plans (scope, schedule, cost, quality, risk, communications, resources, procurement, and stakeholder engagement).
- Direct and Manage Project work
 - Execute tasks according to the plan and coordinate teams.
- Manage Project knowledge
 - Capture lessons learned, maintain documentation, and apply organizational knowledge.

Real world Integration Processes

- Monitor and control project work
 - Track performance using KPIs and dashboards; ensure the project stays on track.
- Perform integrated change control
 - Review and approve change requests to prevent scope creep.
- Close project
 - Formal acceptance of deliverables, release resources, archive documentation.

Menti: Who is responsible for project integration management?

- <https://www.menti.com/alwzaz71hd97>



Change Management

Change Management in IT Projects

Why it matters

- Project success is not only about delivering a technical solution.
- Real value appears when people understand the change, adopt the new way of working, and sustain it.



Project management vs change management

Project management

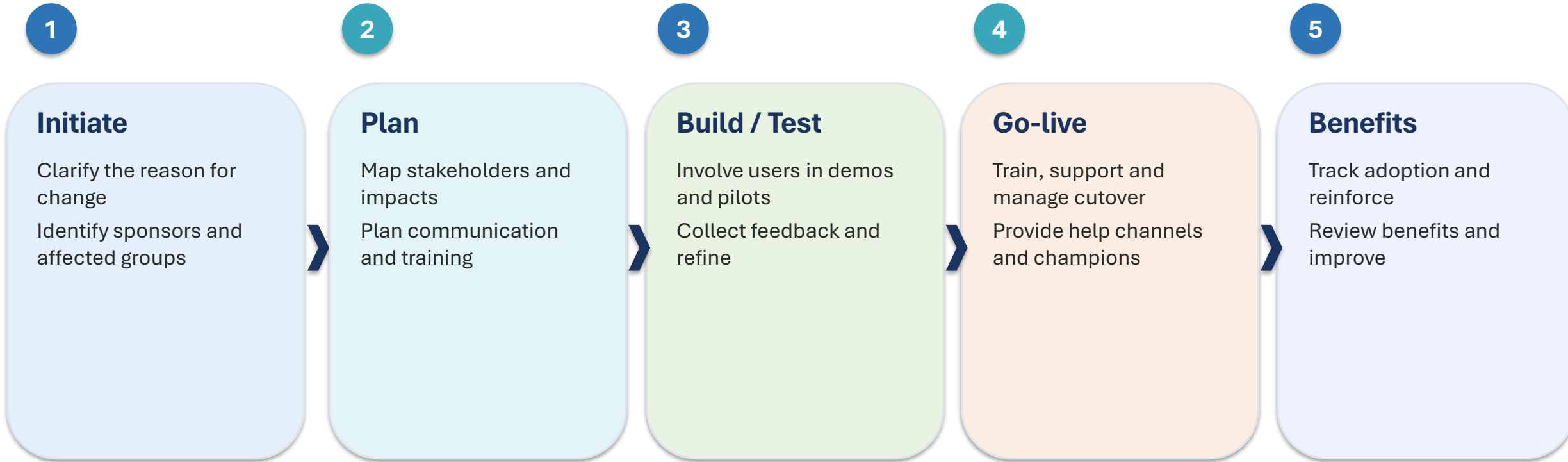
Focuses on scope, schedule, budget and quality
Coordinates resources, vendors and delivery work
Builds and launches the solution
Primary question: “Did we deliver it well?”

Change management

Focuses on communication, engagement and readiness
Builds stakeholder confidence, training and support
Helps people adopt the new way of working
Primary question: “Are people using it successfully?”

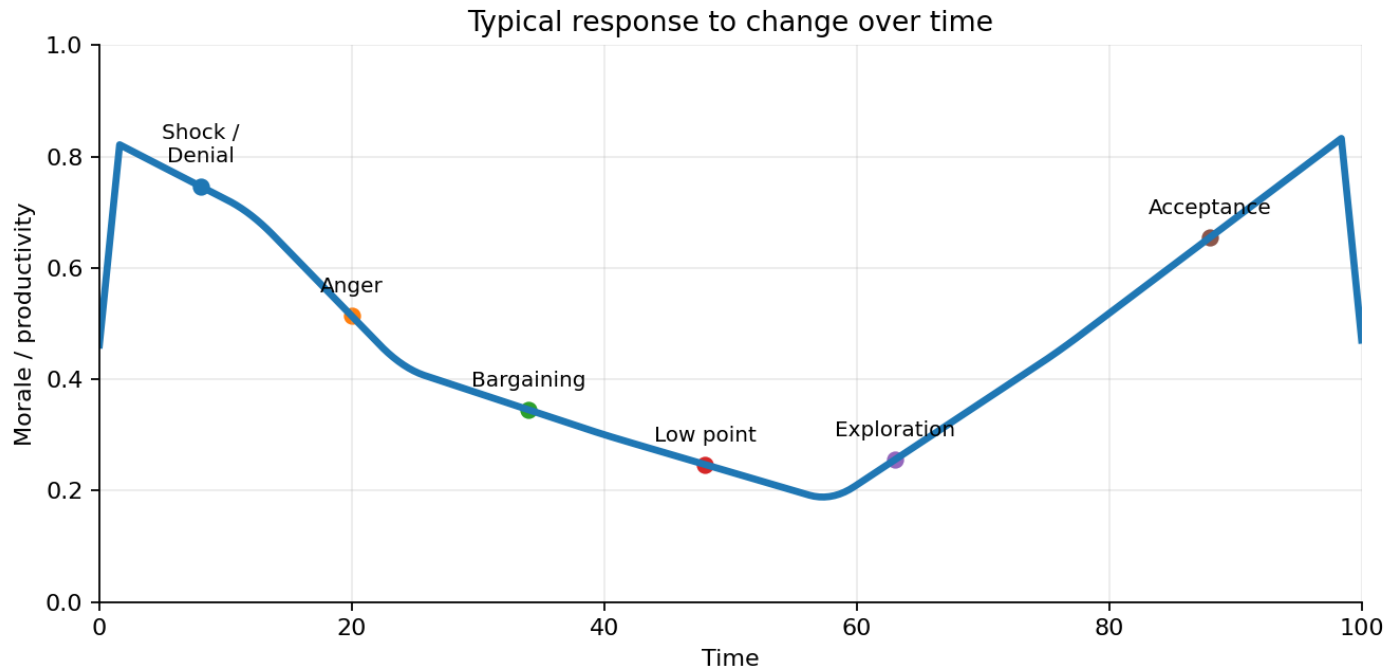
Successful IT change = delivery + adoption

Where change management fits in the project lifecycle



Project manager takeaway: add adoption tasks, owners and measures to the plan - not just technical tasks.

The Kübler-Ross Change Curve



Typical stages: shock/denial → anger → bargaining → low point → exploration → acceptance.

How to use this in projects

- Expect mixed reactions - resistance is normal
- Communicate the “why” early and often
- Listen to concerns and acknowledge uncertainty
- Provide training, job aids and support after go-live
- Look for quick wins to rebuild confidence

Adapted from Kübler-Ross (1969) and workplace change interpretations.

Practical change actions for project managers

Communicate the why

Explain the business reason
Show what changes for users

Map stakeholders

Who is affected?
Who can influence adoption?

Support adoption

Training, job aids, pilots
Managers and champions

Measure uptake

Usage, feedback, compliance
Adjust after go-live

Mini-case: new reporting dashboard rollout

Scenario

A company replaces spreadsheet-based project status reporting with an online dashboard.

The dashboard is technically ready, but team leads continue sending spreadsheet updates because they trust the old process.

Training attendance is low and reporting remains inconsistent across teams.

Discussion prompts

What adoption risk is visible here?

Which stakeholder group needs attention first?

What two change actions would you add to the project plan?

What evidence would show the change is working?

Propose one delivery action and one adoption action.

Key takeaways

- Technical delivery does not guarantee successful change
- Adoption work should start early in the project lifecycle
- Communication, training and support reduce resistance
- Track adoption and benefits after go-live

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- Kotter, J. P. (1996). *Leading Change*. Harvard Business School Press.
- Kübler-Ross, E. (1969). *On Death and Dying*. Macmillan.

For You!

End of Lecture Questions

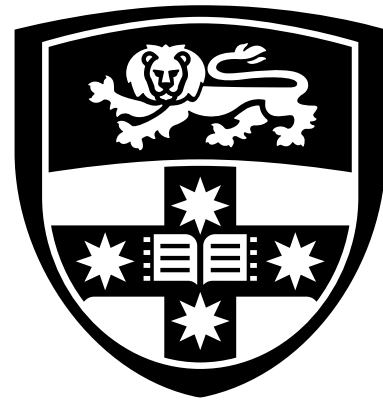
- What is Project Integration Management, and why is it often called the “glue” of project management?
- List the 7 key processes of Integration Management according to PMBOK.
- How does Integration Management help prevent scope creep, cost overruns, and schedule delays?
- How can project knowledge management improve future IT projects?
- In an IT project, which integration process do you think is most likely to fail and why?

Case Study

- EduTech Solutions is launching a new **online learning platform** for university students. The project involves:
 - Developing web and mobile applications.
 - Integrating the platform with the existing university database.
 - Ensuring compliance with data privacy regulations.
 - Coordinating developers, UX/UI designers, IT support, and the university administration.
 - Delivering the platform within 6 months.
- **Challenges:**
 - Midway through the project, a request comes from the university to **add a virtual lab feature**.
 - The integration with the existing database has **performance issues**.
 - Resource availability fluctuates as team members are pulled into other projects.

Case-study based questions

- Which **Integration Management processes** would be most critical at the point when the virtual lab feature is requested? Explain your reasoning.
- What **tools or techniques** could you use to ensure all stakeholders remain informed about changes and progress?
- Map the project tasks described in the case to the **five stages of the project lifecycle** (Initiation, Planning, Execution, Monitoring & Controlling, Closing).
- How would you **engage stakeholders** to ensure the new virtual lab feature is properly prioritized and approved?
- If Integration Management is weak, what specific project failures could occur in this scenario?
- Propose one **lesson learned** that could be documented for future IT projects of similar scope.



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